NMC 2011 O-LEVEL PAPER 2 QUESTIONS

SECTION A : 5 marks each

Qn.1. Without using a calculator, evaluate, $\sqrt{170^2 - 80^2}$.

Qn.2. Nancy usually leaves her cell phone ON. If her cell phone is ON but she is not using it, the battery will last for 24 hours. If she is using it constantly, the battery will last for only 3 hours. Since the last recharge, her cell phone has been ON for 9 hours, and during that time, she used it for 60 minutes. If she does not talk anymore but leaves the cell phone ON, how many more hours will the battery last?

Qn.3. Find the product of all values x which satisfy the equation $16^{x^2+3x-1} = 8^{x^2+3x+2}$.

Qn.4. If $f\left(\frac{2x-1}{2-x}\right) = 2x$, find $f(4)$.

Qn.5. A driver travels from Mbale to Masaka for the first time and keeps stopping to take a closer look at any interesting scenery and ended up driving at an average speed of 50 km/h. On her way back she travels at an average speed of $80km/h$. What was her average speed for the whole journey?

Qn.6. A quadrilateral has these properties: three of its angles are congruent to one another, no set of three vertices are co-linear, and its largest angle is three times as large as its smallest one. What is the value of the smallest angle in degrees?

Qn.7. Hasnah, Robinah, and Jones each took the same 7- question True/False quiz. Hasnah and Robinah each got 6 questions right. Their answers were: $FFTTTTF$ for Hasnah and $TFFTTTF$ for Robinah. When Jones took the same quiz, his answers were $TTFFTTT$. How many answers did Jones get right?

Qn.8. A triangle ABC has sides of lengths: $\overline{AB} = \overline{BC} = \sqrt{3}\ cm$ and $\overline{BM} = \sqrt{(3/4)}\ cm$ where M is the mid point of $\overline{AC}$. Find the value of the smallest angle of triangle ABC and the length of side $\overline{AC}$.

Qn.9. A teacher asks each student in her class to draw a rectangle whose sides are whole numbers and having a perimeter of 502 cm. When marking her students' work, she discovers that the students drew all the different possible rectangles. How many different(distinct) rectangles did the students come with and what is the difference between the largest and smallest possible areas of the rectangles?

Qn.10. In the magic square below, the sum of the numbers in each row, each column, and each diagonal is the same. Find the value of B.

19	A	14
10	B	C
D	E	11

SECTION B : 10 marks each

Qn.11. (a) Find a single percentage discount, that is equivalent to two successive discounts of 10% followed by 20% on a price of a commodity.

(b) How many right angled triangles whose all sides are positive whole numbers, have the property that the area is numerically equal to it's perimeter?

Qn.12. (a) A haunted house has six windows. In how many ways can the ghost enter the house by one window and leave by a different window.

(b) John ordered for 4 pairs of black socks and some additional pairs of blue socks. The price of the black socks per pair is twice that of the blue socks. When the order was received by John, he found out that the number of pairs of the two colors had been interchanged. This interchange led to an increase in the bill of 50%. What is the ratio of the number of pairs of the black socks to the number of pairs of the blue socks in the original order.

Qn.13. (a) Find the distance between the two points where the graph $4x^2 + 4y^2 - 16y = 9$ crosses the y-axis.

(b) Okello travels from point P to point Q in 8 minutes. Mary travels from Q to P along the same route. They start at the same time and each travels at a constant rate. If Mary reaches point P , 18 minutes after they both meet, how many minutes did the entire trip take Mary?

Qn.14. A triangle ABC is such that $\overline{AB} = 20\ cm$, $\overline{AC} = 12\ cm$ and $\angle ACB = 90^0$. A point E on $\overline{CB}$ is a perpendicular bisector of line $\overline{AB}$ and bisects $\overline{AB}$ at a point D. Find the area of the quadrilateral $ADEC$.

Qn.15. (a) Without using a calculator, find the value of $(16)^{0.32} \cdot (256)^{0.09}$?

(b) A rectangle is partitioned into four smaller rectangles, three of which have areas (in cm^2), $15, 21$ and 32 as shown in the figure below. What is the area of the entire rectangle?

15	21
32	

NMC 2010 O-LEVEL PAPER 2 SOLUTIONS

SECTION A

1.

$$\begin{aligned}
\frac{a}{a+b+c+d} &= \frac{4}{3} \\
\Leftrightarrow\ 3a &= 4(b+c+4d \\
\Leftrightarrow\ d &= \frac{3a-4(b+c)}{4} \\
\Leftrightarrow\ \frac{d}{a} &= \frac{3-4\frac{(b+c)}{a}}{4} \\
\Leftrightarrow\ \frac{d}{a} &= \frac{3-4\frac{5}{3}}{4} = -\frac{11}{12}.
\end{aligned}$$

2. The following table shows the position of Bob and Alice on every turn.

Turn	$0th$	$1st$	$2nd$	$3rd$	$4th$	$5th$	$6th$
Bob:	12	3	6	9	12	3	6
Alice:	12	5	10	3	8	1	6

The game will take six turns and they will meet at a pointed marked 6.

3.

$$\begin{aligned}
1+x^2+x^2+x^3+\ldots+x^{2010} &= 1+x^2+x^2+x^3+x^4(1+x^2+x^2+x^3)+\ldots \\
&\quad +x^{2007}(1+x^2+x^2+x^3) \\
&= 0+x^4(0)+\ldots+x^{2007}(0) \\
&= 0.
\end{aligned}$$

4. $f(f(f(x))) = f(f(ax+b)) = f(a(ax+b)+b) = a(a(ax+b)+b)+b = a^3x+a^2b+ab+b.$

$f(f(f(x))) = 8x+21$, it implies that $a^3 = 8 \ \Rightarrow\ a = 2$ and $a^2b+ab+b = 21 \ \Rightarrow\ b = 3$. Therefore, $a+b = 5$.

5. Let the George' original wage be w. After the decrease, his new wage is $w_1 = 75\%w$. If this wage bill is increased by $I\%$, the new wage bill will be $w_2 = 75\% \times (100+I)\%$. Since the new bill is equal to the first bill, we solve for $75\% \times (100+I)\%w = w$ implying that $I = \frac{100}{3}$. Therefore, we have to increase the current wage by 33.33% to get the original wage.

6.
$$\begin{aligned}\left(\sqrt{3}\right)^x \cdot \left(\sqrt{27}\right) &= \sqrt{3} \\ \Rightarrow\ 3^{\frac{x}{2}} \cdot 27^{\frac{x}{2}} &= 3^{\frac{1}{2}} \\ \Rightarrow\ (81)^{\frac{x}{2}} &= 3^{\frac{1}{2}} \\ \Rightarrow\ \left(3^4\right)^{\frac{x}{2}} &= 3^{\frac{1}{2}} \\ \Rightarrow\ (3)^{2x} &= 3^{\frac{1}{2}} \\ \Rightarrow\ 2x &= \frac{1}{2} \ \Rightarrow\ x = \frac{1}{4}.\end{aligned}$$

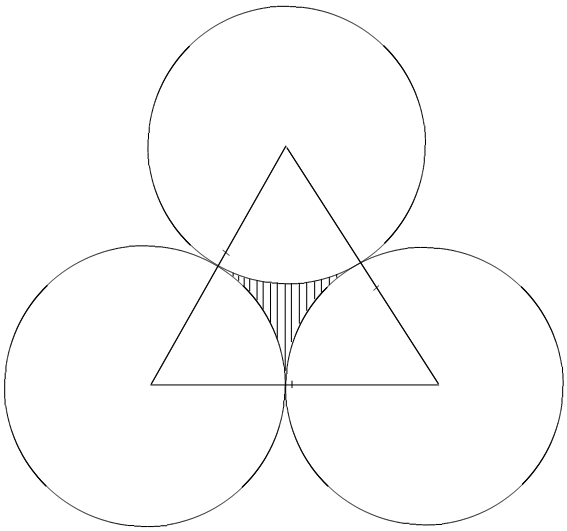
7.
$$\begin{aligned}60 &= x^2 + 3xy + 2y^2 \\ &= x^2 + 2xy + y^2 + xy + y^2 \\ &= (x+y)^2 + y(x+y) \\ &= 36 + 6y.\end{aligned}$$

Solving gives $y = 4$ and $x = 2$. Therefore, $2x + 3y = 16$.

8.
$$\log_{10} x^2 - \log_{10} 2x = 1 \ \Rightarrow\ \log_{10} \frac{x^2}{2x} = 1 \ \Rightarrow \log_{10} \frac{x}{2} = 1 \ \Rightarrow \frac{x}{2} = 10 \ \Rightarrow x = 20.$$

9. Mark's total score on the first 10 quizzes is $85 \times 10 = 850$. Mark's total score on 14 quizzes is $14 \times 90 = 1260$. Therefore, Mark's average score on the last four questions is $\frac{1260-850}{4} = 102.5$

10.



$$\begin{aligned}\text{Area of shaded region} &= \text{Area of the equilateral triangle} - \text{Area of the three sectors}\\ &= \frac{1}{2}\times 2\times 2\times \sin 60 - 3\left(\frac{60}{360}\times \pi \times 1^2\right)\\ &= \sqrt{3}-\frac{\pi}{2}.\end{aligned}$$

11. (a) There are 720 such arrangements. To have an arrangement where the lower case letters appear in an increasing alphabetical order, we note that there are
    - 2 ways in which the caps and lower case letters alternate.
    - $4! = 24$ ways in which abc appear as a block.
    - 36 way in which ab appear as a block, excluding the case abc.
    - 36 ways in which bc appear as a block, excluding the case abc.

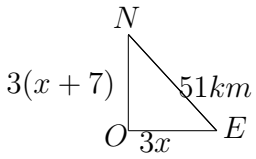
    In all, there are 98 ways in which the lower case letters appear in an increasing alphabetical order. The required probability is therefore, $\frac{98}{720} = 0.136$.

    (b) Let x and y be the length and width of the rectangle. Then, $x^2 + y^2 = 10$ and $xy = 3$. We want to find the perimeter $p = 2(x + y)$.

$$\begin{aligned}(x+y)^2 &= x^2+y^2+2xy\\ &= 10+2\times 3 = 16.\end{aligned}$$

    Therefore, $x + y = \sqrt{16} = 4$ and the perimeter $p = 2(x + y) = 8$.

12. Let Ann's speed be xkm/hr. This implies that Jason's speed is $x + 7km/hr$. If O is the camp site, after three hours, Ann is at point E a distance of $3xkm$ from point o and Jason at point N a distance of $3(x + 7)km$ away from point O. If $\overline{EN} = 51km$, then,



$$\begin{aligned}(3x)^2 + (3(x+7))^2 &= 51^2\\ \Rightarrow\ 9x^2 + 9(x^2 + 14x + 49) &= 51^2\\ \Rightarrow\ 18x^2 + 126x + 441 &= 2601\\ \Rightarrow\ x^2 + 7x - 120 &= 0\ \Rightarrow\ x = 8 \text{ or } -15.\end{aligned}$$

We take $x = 9$ and disregard $x = -15$. The sum of their speed is therefore $x+x+7 = 23km/hr$.

13. Let x denote the time David spends running around the circle. If l is the length of the circular track, the, David's speed is $\frac{l}{x}$ and Juma's speed is $\frac{l}{40}$. When they first meet, the total distance they have traveled is $\frac{l}{40} \times 15 + \frac{l}{x} \times 15 = l$ the length of the track. Solving gives $x = 24$. the average of their time is $\frac{24+40}{2} = 32$ seconds.

14.

	Rainy morning	Non-rainy mornings
Rainy afternoons	a	b
Non-rainy afternoons	c	e

Using the tabulated information, we have,

$$\left.\begin{array}{c} a+b+e+c=d\\ a+b+c=7\\ a=0\\ c+e=5\\ b+e=6 \end{array}\right\} \Rightarrow \left.\begin{array}{c} b+e+c=d\\ b+c=7\\ c+e=5\\ b+e=6 \end{array}\right\} \Rightarrow \left.\begin{array}{c} b+e+c=d\\ b+c=7\\ b-e=2\\ b+e=6 \end{array}\right\}$$

from which we deduce that $b = 4, e = 2, c = 3$ and $d = 9$.

15. Let $\angle DAE = a^0$ and let $\angle EDC = x$. Then,

$$\begin{aligned}\angle ABC + \angle BCA + 30 + a &= 180\\ \Rightarrow\ 2\angle BCA &= 180 - 30 - a\\ \Rightarrow\ \angle BCA &= \frac{180 - 30 - a}{2}\end{aligned}$$

$$= \quad 75 - \frac{a}{2}.$$

Similarly,

$$2\angle DEA + a = 180 \ \Rightarrow \angle DEA = 90 - \frac{a}{2}.$$

Using $\triangle ADC$, we have, $a + 90 - \frac{a}{2} + x + 75 - \frac{a}{2} = 180 \ \Rightarrow \ x = 165 = 180 \ \Rightarrow \ x = 15^0$.